

Mahyar Ghazanfari

George Washington University, Washington, DC, USA, 20052

✉ Mahyar.ghazanfari@gwu.edu — mahyar.ghazanfari@berkeley.edu — mghazanf@umd.edu

 [LinkedIn](#) —  [GitHub](#)

RESEARCH INTERESTS

My research centers on vision-language models (VLMs): how they internally represent visual evidence, how faithfully that representation drives their outputs, and how to deploy them reliably in embodied and safety-critical settings. My core interests are VLM applications in robotics and aerial autonomy, mechanistic and latent-space interpretability of multimodal models, vision-language retrieval, and the reliability of VLM- and LLM-based evaluation. I study visual grounding through causal interventions on internal representations, and I apply the resulting insights to perception, reasoning, and decision support for autonomous drones, self-driving vehicles, and aviation operations. Key topics include multimodal foundation models, interpretability and AI safety, image-text retrieval, agentic LLM pipelines, and human-AI collaboration in high-stakes environments.

EDUCATION

Ph.D. in Mechanical and Aerospace Engineering

George Washington University, School of Engineering and Applied Sciences

Advisor: Dr. Peng Wei — Research: *Vision-language models, multimodal interpretability, and safety-critical decision making*

September 2024 – Present

Washington, DC, USA

GPA: 4.0/4.0

Visiting Graduate Student, Department of Computer Science

University of Maryland, College Park

Graduate coursework and research in *computer vision, vision-language models, and multimodal learning*

September 2025 – January 2026

College Park, MD, USA

B. S. in Electrical Engineering

Sharif University of Technology

Advisors: Dr. Babak Khalaj, Dr. Ali Fotowat — Research: *Sensor Fusion, Control and Path Planning, Model Predictive Control (MPC)*

September 2019 – June 2024

Tehran, Iran

GPA: 17.18/20

PUBLICATIONS

Accepted / Published

1. **Mahyar Ghazanfari**, Peng Wei. *A Paragraph is Worth a Thousand Captions: Rethinking Text Supervision for Long Caption Retrieval*. Accepted at the **Workshop on Multimodal Understanding and Cross-modal Grounding (MUCG)**, ECCV, 2026.
2. **Mahyar Ghazanfari**, Amin Tabrizian, Armin Mehrabian, Peng Wei. *EO-Agents: A Three-Agent LLM Pipeline for Earth Observation Hypothesis Generation*. Accepted at the **AI for Science (AI4Science) Workshop**, ICML, 2026.
3. Amin Tabrizian, Arsyi Aziz, **Mahyar Ghazanfari**, Peng Wei. *End-to-End LLM Flight Planning with RAG-based Memory and Multi-modal Coach Agent*. Accepted at the **Workshop on Language Models for Planning (LM4Plan)**, ICML, 2026.
4. **Mahyar Ghazanfari**, Iman Sharifi, Peng Wei, Abenezer Taye, Bryan Ward, Xenofon Koutsoukos, Gautam Biswas, Mahsa Ghasemi, Vijay Gupta, Hyeong Tae Kim, et al. *A Survey of Security Challenges and Solutions for Advanced Air Mobility and eVTOL Aircraft*. In **AIAA SCITECH 2026 Forum**, 2026.
5. Iman Sharifi, **Mahyar Ghazanfari**, Abenezer Taye, Peng Wei, Maheed Ahmed, Hyeong Tae Kim, Mahsa Ghasemi, Vijay Gupta, Noah W. Dahle, Robert Canady, et al. *A Survey of Security Challenges and Solutions for UAS Traffic Management (UTM) and Small Unmanned Aerial Systems (sUAS)*. In **AIAA SCITECH 2026 Forum**, 2026.
6. Amin Tabrizian, **Mahyar Ghazanfari**, Peng Wei. *Chain-of-Thought Flight Planner: End-to-End LLM Routing Under Wind Hazards*. In **AIAA Aviation Forum and ASCEND**, 2025. (Best Paper Award)

7. Can X. Vu, **Mahyar Ghazanfari**, Kevin Dong, Abenezer Taye, Amin Tabrizian, Peng Wei. *Transformer or CNN? Benchmarking Real-Time Detection Transformer and YOLOv8 for Small UAS Autonomous Landing*. In **AIAA Aviation Forum and ASCEND**, 2025.
8. **Mahyar Ghazanfari**, Peng Wei. *See and Say: Vision-Language Guided Safe Zone Detection for Autonomous Package Delivery Drones*. Accepted at the **IEEE/AIAA Digital Avionics Systems Conference (DASC)**, 2026.
9. T. Darrell, **Mahyar Ghazanfari**, J. Kam, A. Bayen, A. Tabrizian, D. Vu, P. Wei. *Automated Multimodal Analysis of Air Traffic around Non-Towered Airports using Large Language Models*. Accepted at **AIAA Aviation Forum**, 2026.

Under Review

10. **Mahyar Ghazanfari**, Milad Heirdarisorkhedizaj, Peng Wei. *Looking Without Seeing: Visual Grounding Is Frozen at the Prompt in Vision-Language Models*. Under review at the **AAAI Conference on Artificial Intelligence (AAAI)**, 2027.

In Preparation / Ready for Submission

11. **Mahyar Ghazanfari**, Amin Tabrizian, Arsyi Aziz, April Yang, Peng Wei. *BRAKE: A Benchmark for Road Anomaly Knowledge and Explanation in Driving Scenes*. To be submitted to **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, 2027.
12. **Mahyar Ghazanfari**, Peng Wei. *VLM Judges Cannot Judge Better Than They Can Understand: On the Perception Ceiling of Vision-Language Judges*. To be submitted to the **International Conference on Learning Representations (ICLR)**, 2027.

RESEARCH EXPERIENCE

Visiting Researcher

May 2026 – August 2026

Berkeley Artificial Intelligence Research (BAIR) Lab, University of California, Berkeley

- Visiting researcher hosted by **Prof. Alexandre Bayen** and **Prof. Trevor Darrell**.
- Conducted research on **mechanistic interpretability of vision-language models**, using causal attention knockouts and state-corruption tracing to localize where and when visual grounding is established during inference.
- Studied **vision-language retrieval and latent-space structure** of multimodal representations, including the role of long-form text supervision in image-text alignment.
- Contributed to **LLM-based multimodal analysis of air traffic** around non-towered airports, integrating radio communication transcripts with trajectory and weather data.

Graduate Research Assistant

September 2024 – Present

Intelligent Aerospace Systems Lab (IASL), George Washington University

- Conduct research supported by the **NASA University Leadership Initiative (ULI)** on applying large language models (LLMs) and vision-language models (VLMs) to safety-critical aviation systems.
- Investigate **VLM interpretability and reliability**, including causal analysis of visual grounding, latent-space probing of multimodal representations, and the validity of VLM-as-judge evaluation protocols.
- Develop **LLM-based flight planning and routing algorithms** that incorporate wind hazards and natural-language reasoning for advanced air mobility and eVTOL operations.
- Design a **multimodal VLM framework for non-towered airport safety analysis** that fuses CTAF radio transcripts, ADS-B aircraft trajectories, METAR weather reports, and VFR sectional charts to automatically detect unsafe communication patterns and generate safety advisories.
- Build the **See&Say VLM-guided perception framework** for autonomous drone delivery, combining monocular depth estimation, open-vocabulary detection, and VLM-based reasoning to generate safety maps and identify alternative package drop zones in dynamic urban environments.

- Develop end-to-end pipelines that integrate **vision, language, and structured aviation data** to improve situational awareness, decision support, and human–AI collaboration in safety-critical aerospace operations.

Research Assistant

June 2023 – June 2024

Advanced Integrated Circuit Design (AICD) Lab, Sharif University of Technology

- Conducted research on **sensor fusion for autonomous driving** using the CARLA simulator as a high-fidelity testing environment.
- Developed multimodal fusion pipelines combining **LiDAR, IMU, and GNSS** data using classical filtering approaches including **UKF and EKF**.
- Investigated **deep learning–based multimodal perception** methods to improve environment understanding in self-driving vehicles.
- Explored the **Meta-Transformer framework** and other state-of-the-art architectures for cross-modal representation learning.
- Studied and implemented **state-of-the-art perception and fusion techniques** for robust and reliable autonomous vehicle systems.

Research and Development Intern

July 2023 – Sep 2023

Iran Khodro Powertrain Company (IPCO)

- Contributed to hybrid and electric vehicle projects including **Battery Management Systems (BMS)** and passive battery charging.
- Performed **Hazard Analysis and Risk Assessment (HARA)** and statistical analysis of vehicle component degradation.

NLP Research Intern

June 2022 – Aug 2022

ASR Gooyesh Pardaz

- Developed and trained **natural language processing and speech-related models**.
- Worked on **data preprocessing, feature extraction, and neural network training** to improve audio and language processing systems.

TEACHING EXPERIENCE

Teaching Assistant

2022 – 2024

Department of Electrical Engineering, Sharif University of Technology

- **Introduction to Machine Learning** — Designed programming assignments and supported students in Machine Learning concepts and implementation .
- **Convex Optimization** — Developed computer assignments and assisted students with optimization theory and numerical methods .
- **Mathematical Methods in Engineering** — Graded homework and quizzes and supported course logistics .
- **Modern Control Systems** — Graded coursework and assisted with course materials .
- **Electronics II** — Graded exams and quizzes and supported course administration .

PROFESSIONAL SERVICE

Journal Reviews

- **Journal of Aerospace Information Systems (JAIS)** — 4 papers (2024–2026)

Conference Reviews

- **IEEE International Conference on Robotics and Automation (ICRA)** — 1 paper (2025)
- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)** — 1 paper (2026)

Workshop Reviews

- **SCALE Workshop, ICML 2026** — 4 papers
- **Privacy, Fairness, Accountability and Transparency in Computer Vision (PFATCV) Workshop, ECCV 2026** — 4 papers
- **Multimodal Understanding and Cross-modal Grounding (MUCG) Workshop, ECCV 2026** — 3 papers
- **Explainable Computer Vision (eXCV) Workshop, ECCV 2026** — 1 paper

TECHNICAL SKILLS

Programming	Python, Java, MATLAB/Simulink, C/C++, Verilog
ML/AI Libraries	PyTorch, JAX, TensorFlow, Hugging Face Transformers, NumPy, Scikit-Learn, OpenCV, Pygame, Ollama Framework
Software Tools	HSPICE, PSPICE, Proteus, Git, L ^A T _E X

AWARDS AND FELLOWSHIPS

Best Paper Award, AIAA Aviation Forum Summer 2025
For the paper *Chain-of-Thought Flight Planner: End-to-End LLM Routing Under Wind Hazards*, Las Vegas, NV.

NASA University Leadership Initiative (ULI) GRA Award 2024 – Present
George Washington University Graduate Research Assistantship

Top 1% Rank, Iranian National University Entrance Exam (Konkur) 2019
Ranked within the top 1% among more than 300,000 participants in the national exam for admission to undergraduate studies.

SELECTED COURSEWORK

Machine Learning (A)	Deep Reinforcement Learning (A)	Robotic Systems (A)
Model Predictive Control (A)	Electro-Mechanical Control Systems (A)	Communication Systems (A)
Probability and Statistics (A)	Robotics and Computer Vision (A)	Industrial Control (A)
Digital Control (A)	Multimodal Foundation Models (A)	Large Language Vision Models (A)

SERVICE AND LEADERSHIP

- **Workshop Organizer** — Helped organize and host two NASA ARMD System-Wide Safety (SWS) workshops at George Washington University: *In-Time Safety Management for Emerging Aviation Operations* and *Assurance of Autonomy* April 2025
- **Workshop Organizer** — Helped organize the *Safe and Reliable AI Workshop* at George Washington University with participants from academia, industry, and government agencies October 2024
- **NASA ULI Program Workshop Organizer** — Helped organize the NASA University Leadership Initiative workshop with collaborators from NASA, MIT Lincoln Laboratory, Purdue University, UC Irvine, Vanderbilt University, and Collins Aerospace October 2024
- **Research Mentor** — Mentored several high school students on research in autonomous UAV perception and computer vision May 2025
- **Member, RASTA Student Organization** — Contributed to designing workshops introducing advanced concepts in probability and statistics for high school students January 2022

REFERENCES

Dr. Peng Wei

Associate Professor, Department of Mechanical and Aerospace Engineering
The George Washington University, Washington, DC, USA.
pwei@gwu.edu

Prof. Alexandre Bayen

Liao-Cho Professor of Engineering; Professor, Department of Electrical Engineering and Computer Sciences and Department of Civil and Environmental Engineering

University of California, Berkeley, Berkeley, CA, USA.

bayen@berkeley.edu

Prof. Qi Alfred Chen

Assistant Professor, Department of Computer Science

University of California, Irvine, Irvine, CA, USA.

alfchen@uci.edu

Prof. Ali Fotowat

Associate Professor, Department of Electrical Engineering

Sharif University of Technology, Tehran, Iran

afotowat@sharif.edu